CONDITION OF TEETH AT DIFFERENT AGES

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INTENDED LEARNING OUTCOMES (ILO'S)

- ✓ Define the Chronology of teeth.
- ✓ Review the chronology of Primary dentition.
- ✓ Review the chronology of Permanent dentition.
- ✓ Determine the condition of teeth at different ages.
- ✓ Determine the dental age of a patient based on radiographs and diagrams.

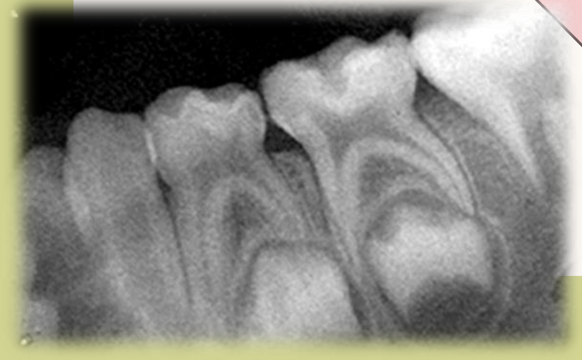
CHRONOLOGY

Chrono: Time

Logy: Study

SO, CHRONOLOGY IS DEFINED AS:

"A STUDY THAT DEALS WITH THE TIMINGS OF VARIOUS STAGES OF TOOTH DEVELOPMENT, STARTING FROM THE INITIATION OF TOOTH CALCIFICATION TILL THEIR ERUPTION IN THE ORAL CAVITY



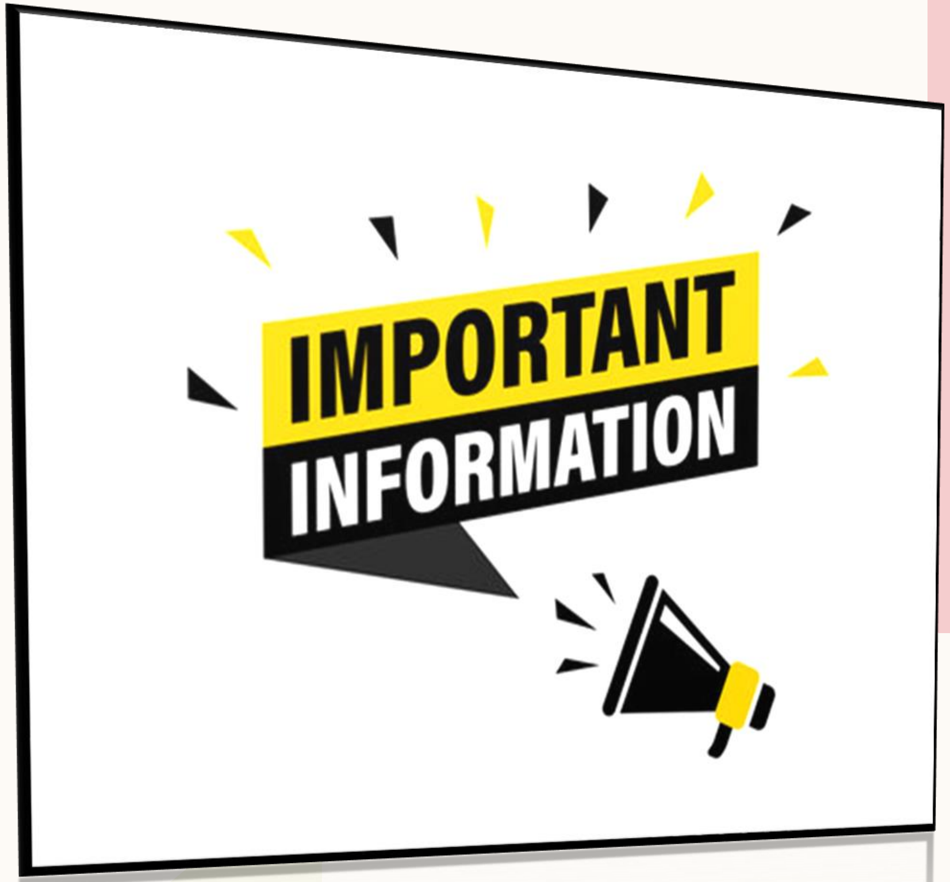
1- Teeth begin their calcification long before their date of eruption.

2-**<u>Eruption</u>** means the *emergence* of the tooth into the oral cavity.

3- When the tooth erupts, *one-third* of the root is completed.

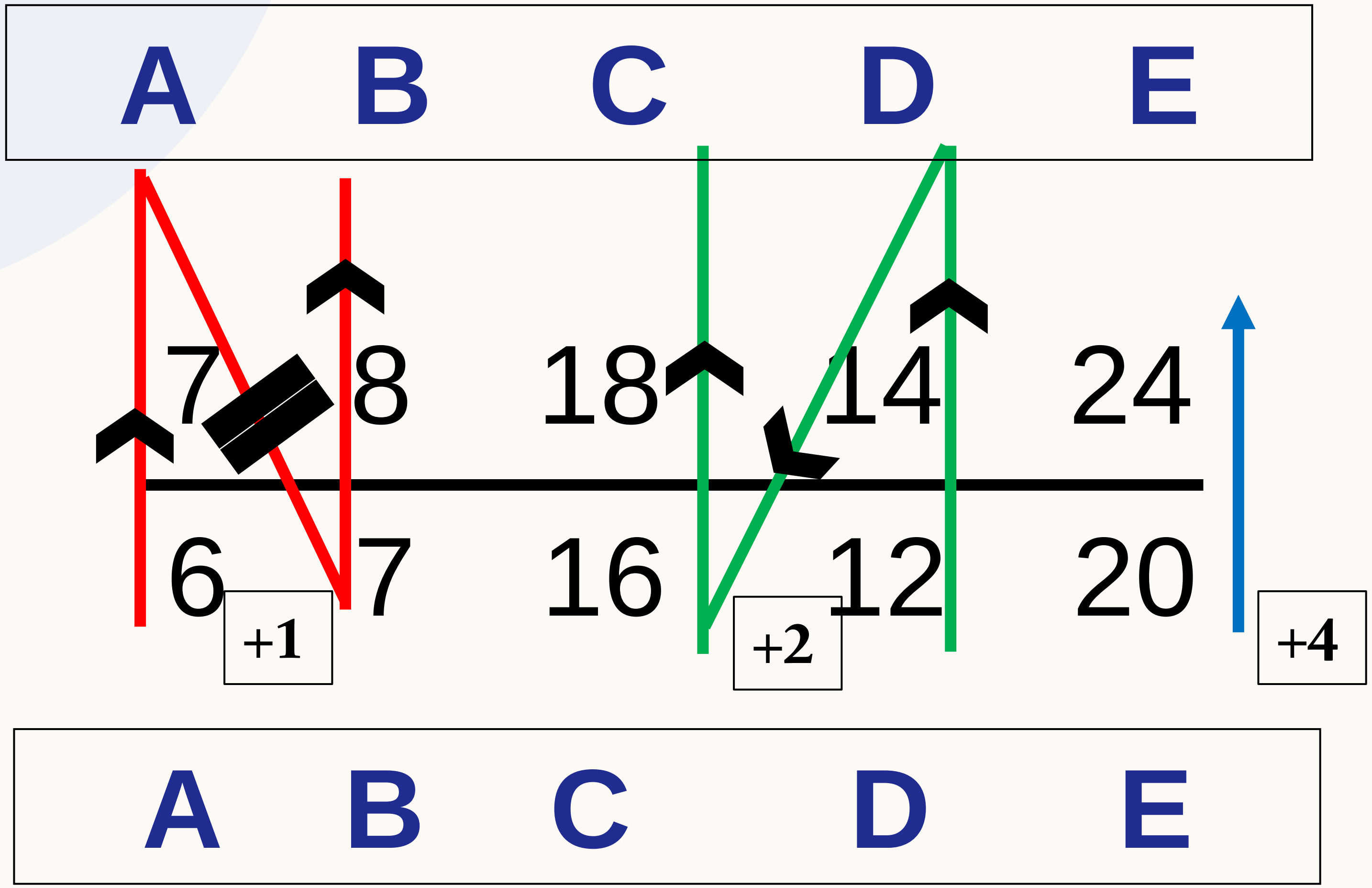
4- Root formation is completed after eruption by *1-3 years*.

5- Roots of deciduous teeth begin to resorb *3 years* before shedding.



CHRONOLOGY OF PRIMARY TEETH

	Calcification Date (m.i.u)	Eruption (months)	Root completed (Years)	Beginning of root resorption
A / A	4	7 / 6	1.5	~ 3 years before shedding
B / B	4.5	8 / 7		
C / C	5	18 / 16	3	
D / D		14 / 12	2.5	
E / E	6	24 / 20	3	


A B C D E
7 8 18 14 24
6 7 16 12 20
+1
+2
+4
A B C D E

SEQUENCE OF ERUPTION OF PRIMARY TEETH

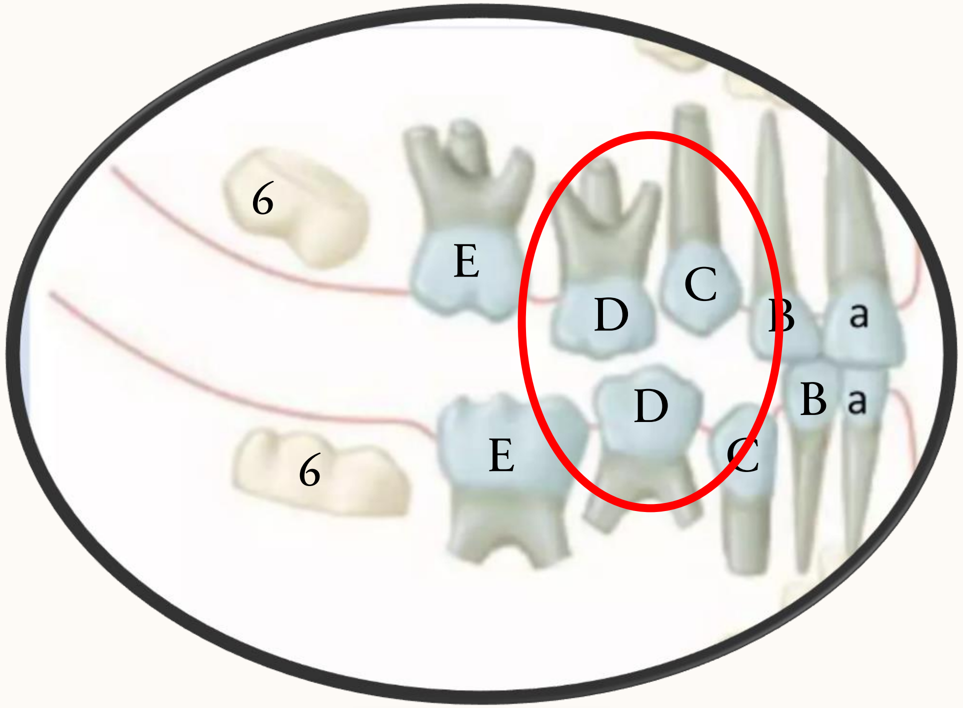
A

B

D

C

E



IMPORTANT CHRONOLOGICAL FACTS ABOUT PRIMARY DENTITION

- Enamel organs for deciduous teeth appear from ***7-9 w.i.u.***
- Beginning of calcification starts from ***4-6 m.i.u.***
- Canines erupt ***after*** primary 1st molars (**D**).
- Mandibular teeth generally emerge ***before*** maxillary teeth.
- Roots are completed after ***1 year*** of eruption for incisors.
- Roots are completed after ***1.5 years*** for canines& molars.
- Roots start resorption at ***3 years*** before shedding.

Finally, Shedding occurs on the same date as the eruption of successors.

SHEDDING (EXFOLIATION)

It is the *physiological elimination* of the deciduous tooth at specific ages due to the *resorption of their roots* before the eruption of their permanent successor.

CHRONOLOGY OF PERMANENT ANTERIOR

	Calcification date (Months)	Crown completed	Eruption (Years)	Root completed
1/1	3-4	- 3 years before eruption	7/6	+ 3 years after eruption
2/2	10-12/3-4		8/7	
3/3	4-5		11/9	

CHRONOLOGY OF PREMOLARS

	Calcification date	Crown completed	Eruption *(Years)*	Root completed
$\frac{4}{4}$	$\frac{1\frac{1}{2}}{1\frac{3}{4}}$	*- 3 years before eruption*	10	*+ 3 years after eruption*
$\frac{5}{5}$	$\frac{2}{2\frac{1}{4}}$		11-12	

CHRONOLOGY OF PERMANENT MOLARS

	Calcification date	Crown completed	Eruption *(Years)*	Root completed
$\frac{6}{6}$	*At birth*		6	
$\frac{7}{7}$	**3** years	*-3-4 years before eruption*	12	*+ 3 years after eruption*
$\frac{8}{8}$	**8** years		18	

+ 6 years

1 2 3 4 5 6 7 8

+1 +1 +6

7 8 **11** 10 11-12 6 12 18

6 7 9 10 11-12 6 12 18

1 2 3 4 5 6 7 8

SEQUENCE OF ERUPTION OF PERMANENT TEETH

Maxillary Arch

6
1
2
4
3
5
7
8

Mandibular Arch

6
1
2
3
4
5
7
8

DENTAL CONDITION AT BIRTH

No teeth are seen in the oral cavity

- Initial mineralization of all primary teeth is **PRENATAL.**
- 1st **molar** is the only permanent tooth to mineralize prenatally.

Deciduous teeth:

Parts of A, B, cusp tip of C, occlusal third of D and cusp tips of E are formed.

Permanent teeth:

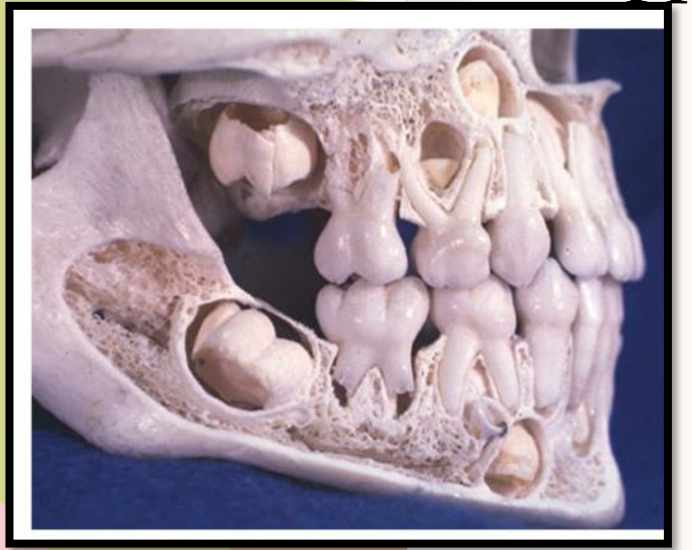
The mesiolingual cusp of upper 6 and lower 6 are formed.

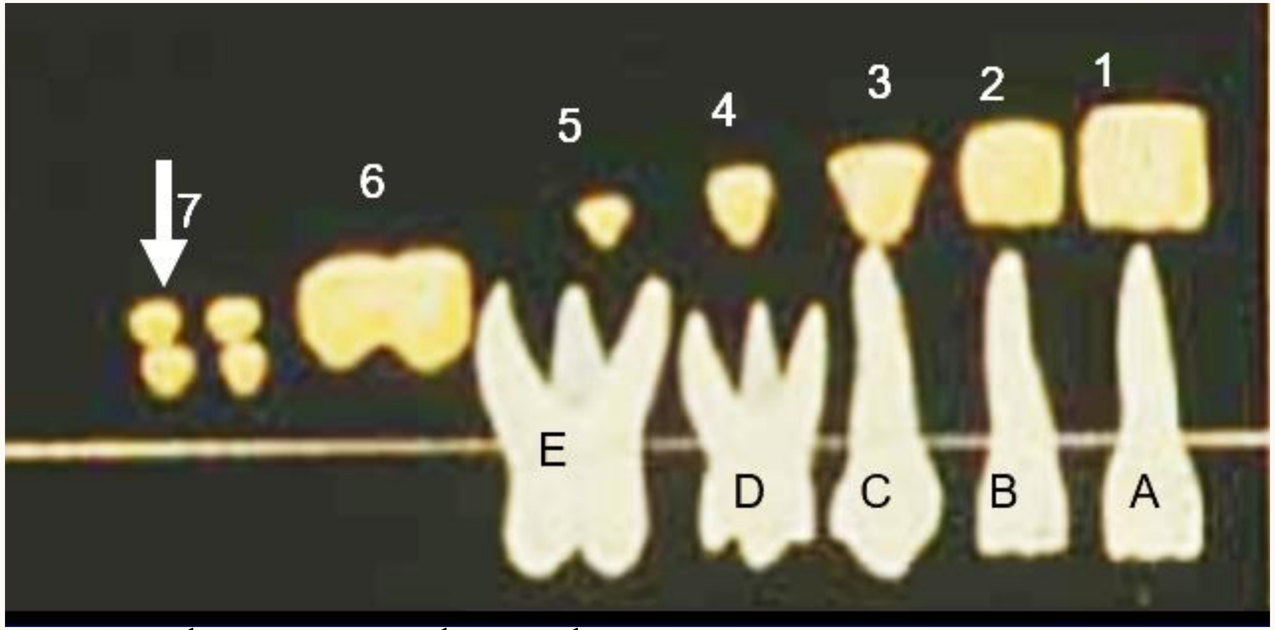
DENTAL CONDITION AT 3 YEARS

Deciduous teeth:

All deciduous teeth are erupted with fully formed roots at about 3-3.5 years.

(The upper and lower teeth are in occlusion)





Permanent teeth:

-Crowns of 6 and 1 are nearly completed.

-Variable parts of 2, 3, 4, and 5 are formed.

-Crown of 7 begins to calcify so separate cusps can be seen.

DENTAL CONDITION AT 6 YEARS

Deciduous teeth:

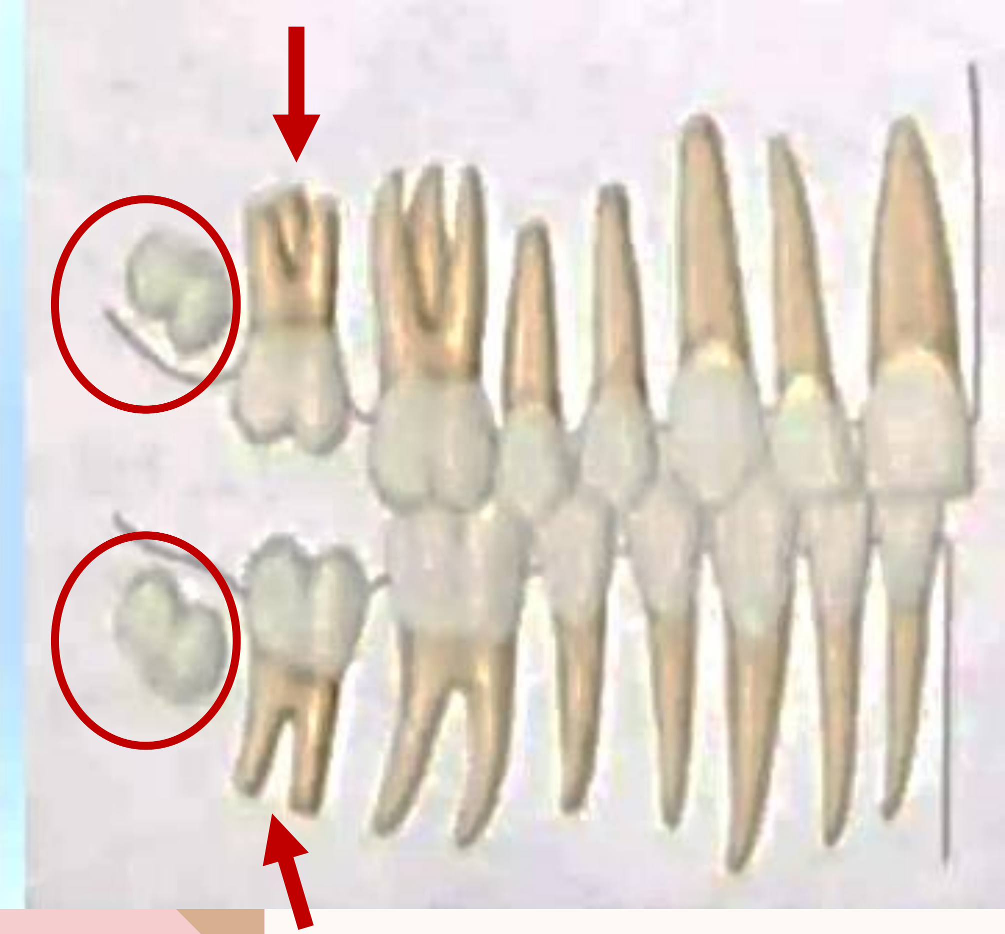
- A is about to be shed and replaced by 1 (especially lower 1).
- B and D show different amounts of root resorption.
- C and E do not present any root resorption.

Permanent teeth:

- Lower 1 is about to be erupted and has 1/3 of its root is formed.
- Crowns of 2 and upper 1 are completed as well as short parts of their roots.
- Crowns of 4 and lower 3 may be completed.
- The crowns of 5, 7 and upper 3 are not yet completed.
- 6 is erupted with 1/3 of its root is formed.

DENTAL CONDITION AT 12 YEARS

All Primary teeth are Shed or Exfoliated.



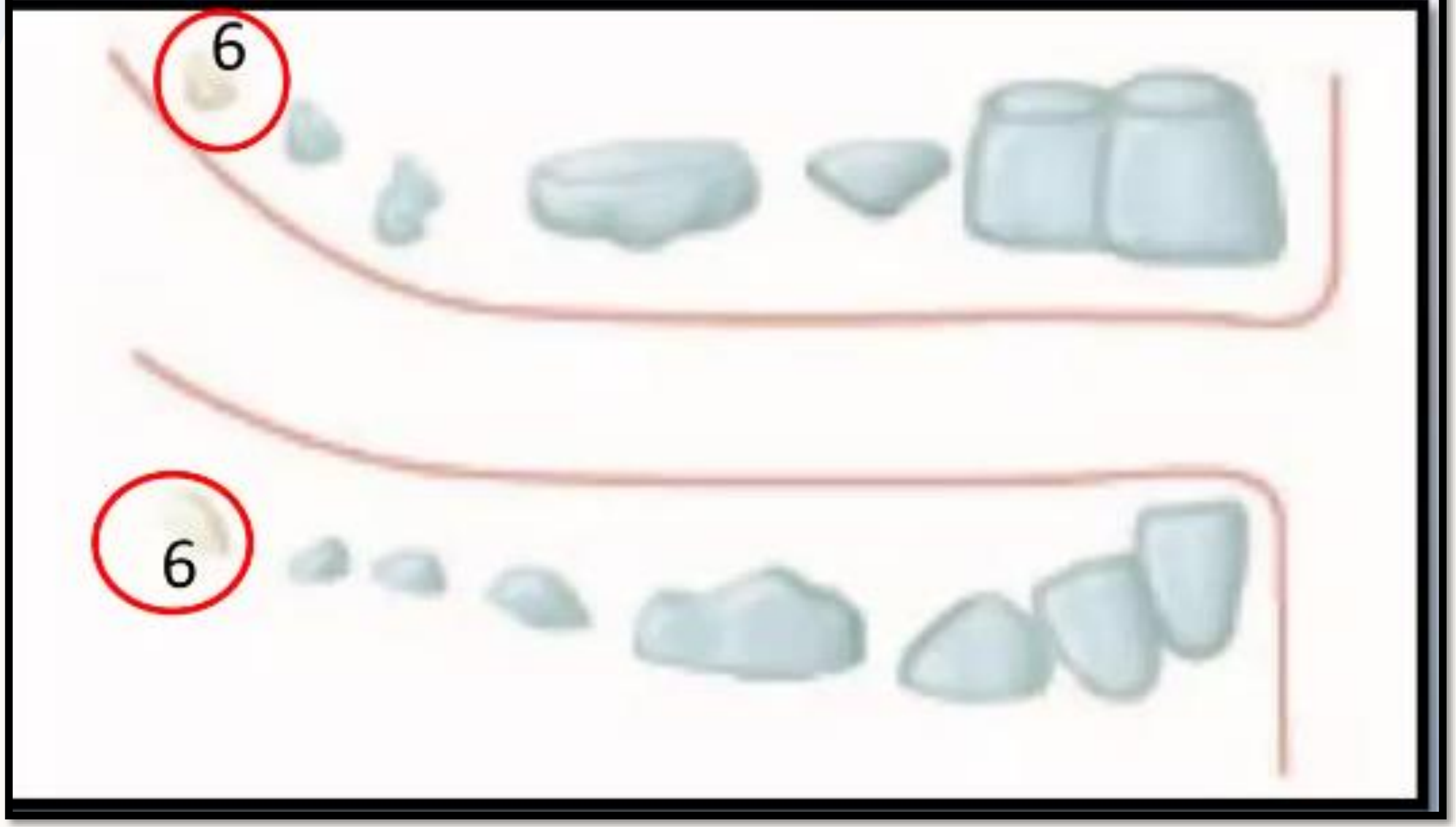
- All permanent teeth erupted except 8.
- All roots are completed except upper 3 & 4&5&7&8.

WHAT IS THE APPROPRIATE DENTAL AGE???

	Calcification Date (m.i.u)	Eruption (months)	Root completed (Years)	Beginning of root resorption
A/A	4	7/6	1.5	- 3 years before shedding
B/B	4.5	8/7		
C/C	5	18/16	3	
D/D		14/12	2.5	
E/E	6	24/20	3	

	Calcification date (Months)	Crown completed	Eruption (Years)
1/1	3-4	- 3 years before eruption	7/6
2/2	10-12 / 3-4		8/7
3/3	4-5		11/9

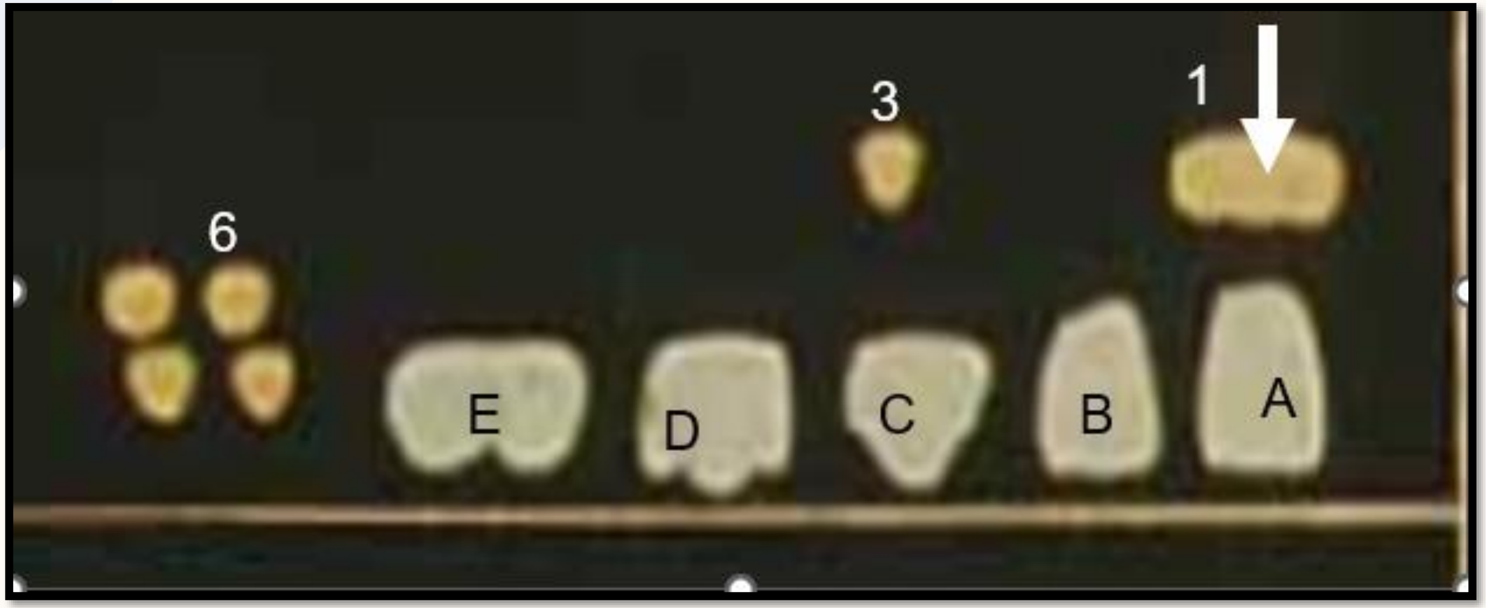
Due to calcification 6 – No successor calcification appeared as at 3-4 months as upper 1 begin calcification.



AT BIRTH – 3 MONTHS

	Calcification Date (m.i.u)	Eruption (months)	Root completed (Years)	Beginning of root resorption
A / A	4	7 / 6	1.5	- 3 years before shedding
B / B	4.5	8 / 7		
C / C	5	18 / 16	3	
D / D		14 / 12	2.5	
E / E	6	24 / 20	3	

	Calcification date (Months)	Crown completed	Eruption (Years)
1 / 1	3-4	- 3 years before eruption	7 / 6
2 / 2	10-12 / 3-4		8 / 7
3 / 3	4-5		11 / 9

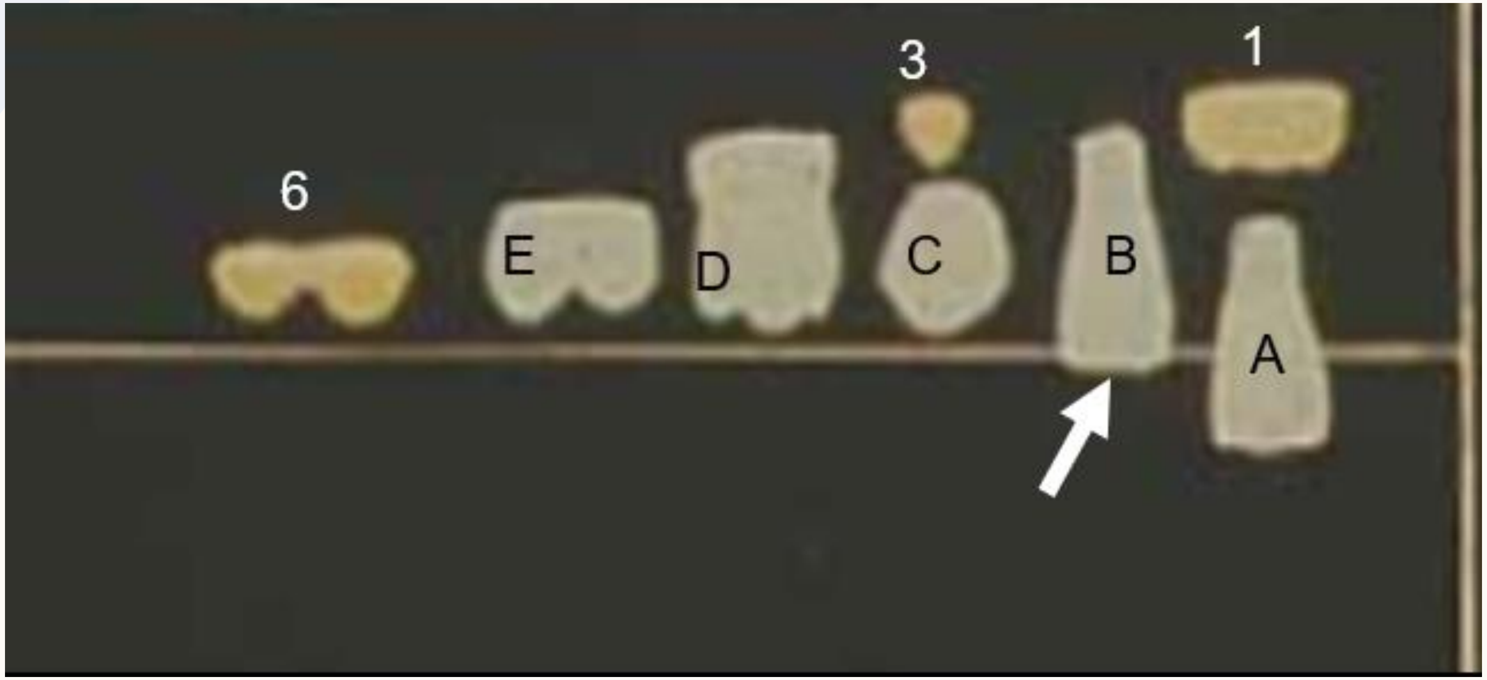


Since calcification of upper 3 is at 4-5 months –
Before the eruption date of upper A.

5-7 MONTHS

	Calcification Date (m.i.u)	Eruption (months)	Root completed (Years)	Beginning of root resorption
A/A	4	7/6	1.5	- 3 years before shedding
B/B	4.5	8/7		
C/C	5	18/16	3	
D/D		14/12	2.5	
E/E	6	24/20	3	

	Calcification date (Months)	Crown completed	Eruption (Years)
1/1	3-4	- 3 years before eruption	7/6
2/2	10-12 / 3-4		8/7
3/3	4-5		11/9



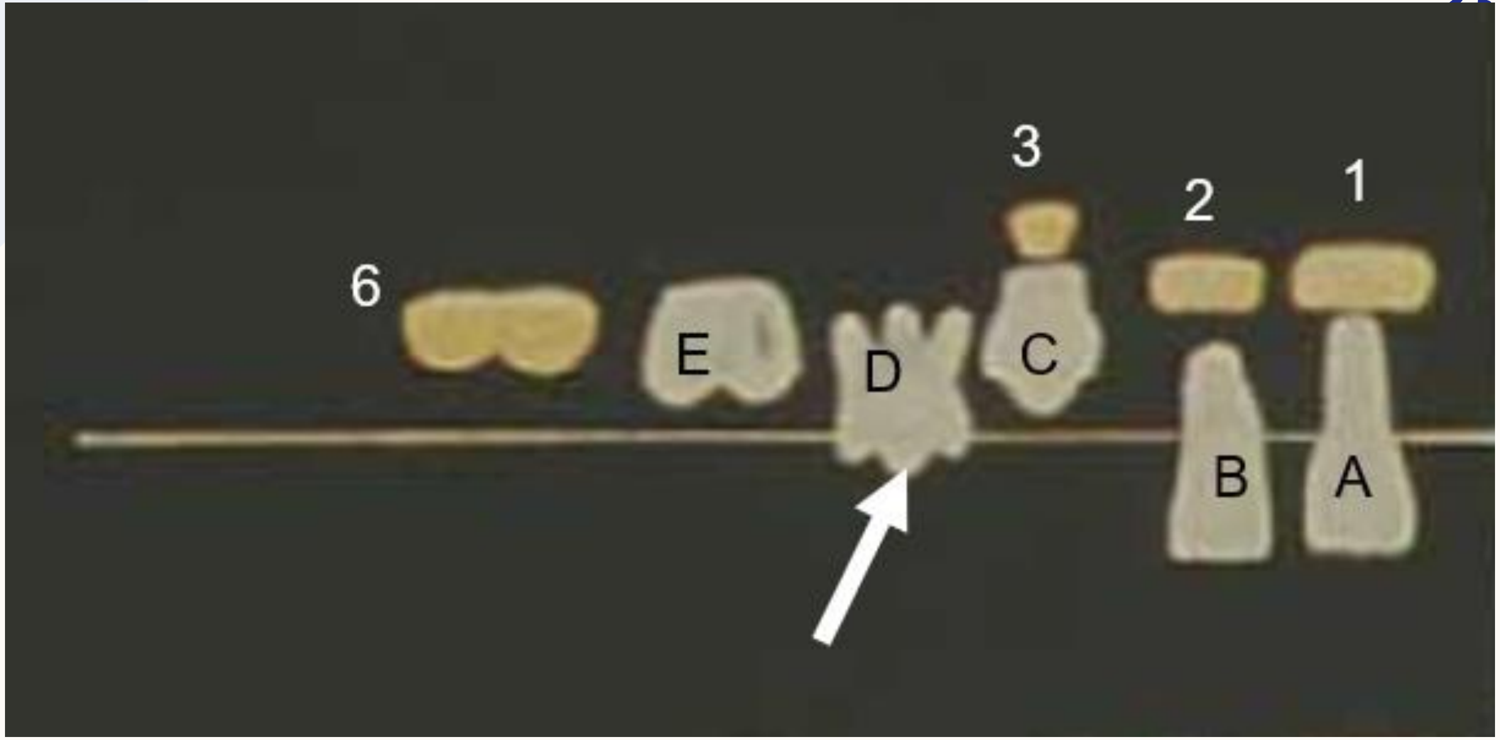
8-10 MONTHS

Since 8 months is the eruption date of upper B – No calcification of upper 2 which begin at 10-12 months.

	Calcification Date (m.i.u)	Eruption (months)	Root completed (Years)	Beginning of root resorption
A/A	4	7/6	1.5	- 3 years before shedding
B/B	4.5	8/7		
C/C	5	18/16	3	
D/D		14/12	2.5	
E/E	6	24/20	3	

4/4	1½ / 1¾	- 3 years before eruption	10
5/5	2 / 2¼		11-12

Due to the beginning of eruption of upper D – Till eruption of upper C at 18 months AND also beginning of calcification of upper 4 at 18 months.

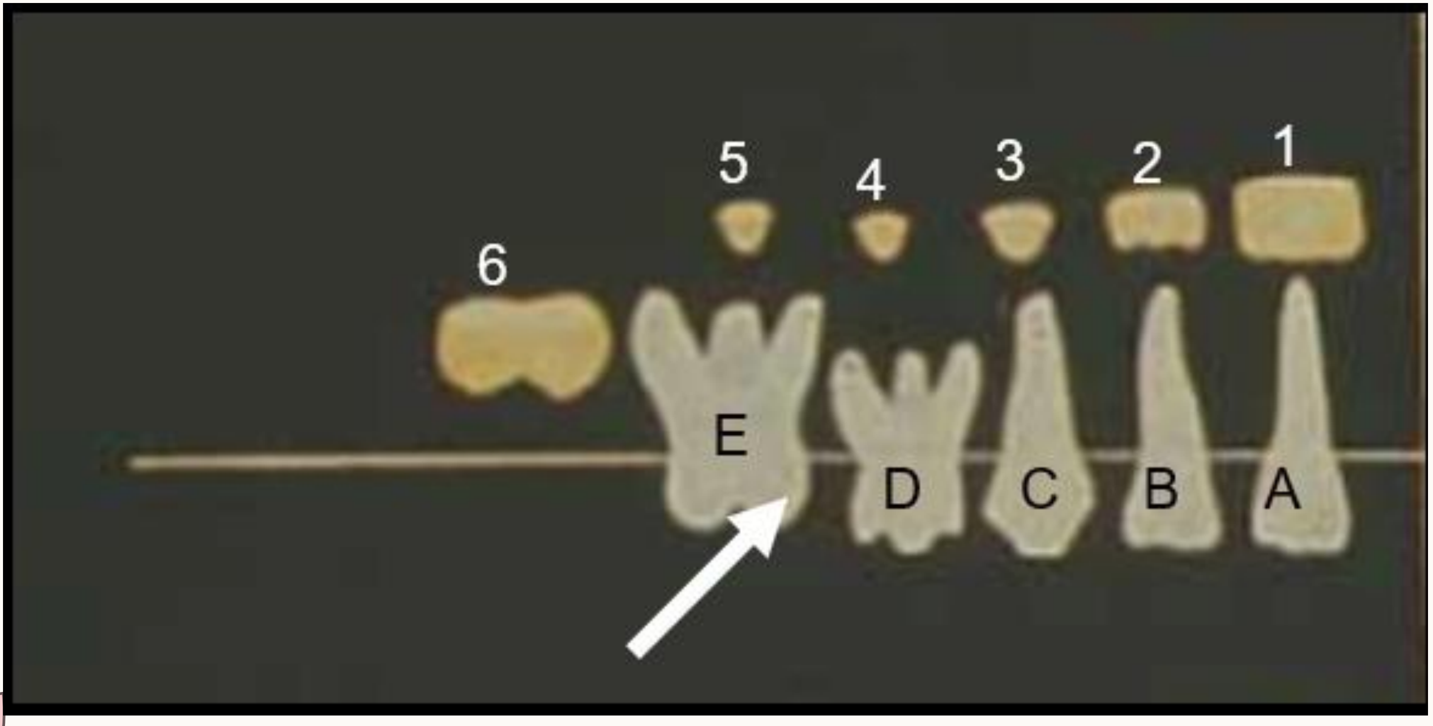


14-18 MONTHS

	Calcification Date (m.i.u)	Eruption (months)	Root completed (Years)	Beginning of root resorption
A/A	4	7/6	1.5	- 3 years before shedding
B/B	4.5	8/7		
C/C	5	18/16	3	
D/D		14/12	2.5	
E/E	6	24/20	3	

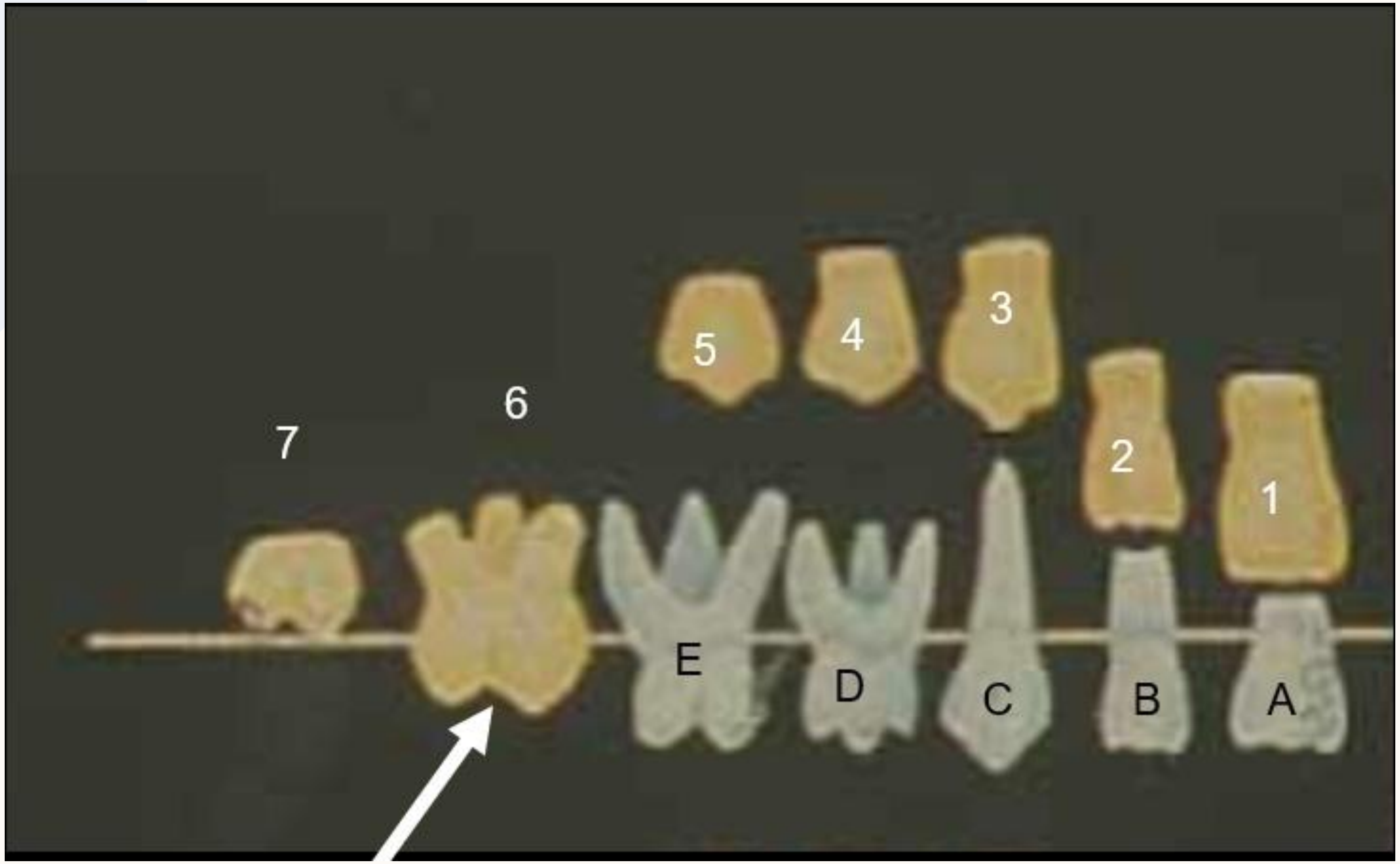
6/6	At birth	-3 -4 years before eruption	6
7/7	3 years		12
8/8	8 years		18

Due to eruption of upper E at 2 years–
Till calcification of upper 7 at 3 years AND roots of upper C and E not completed.



2-3 YEARS

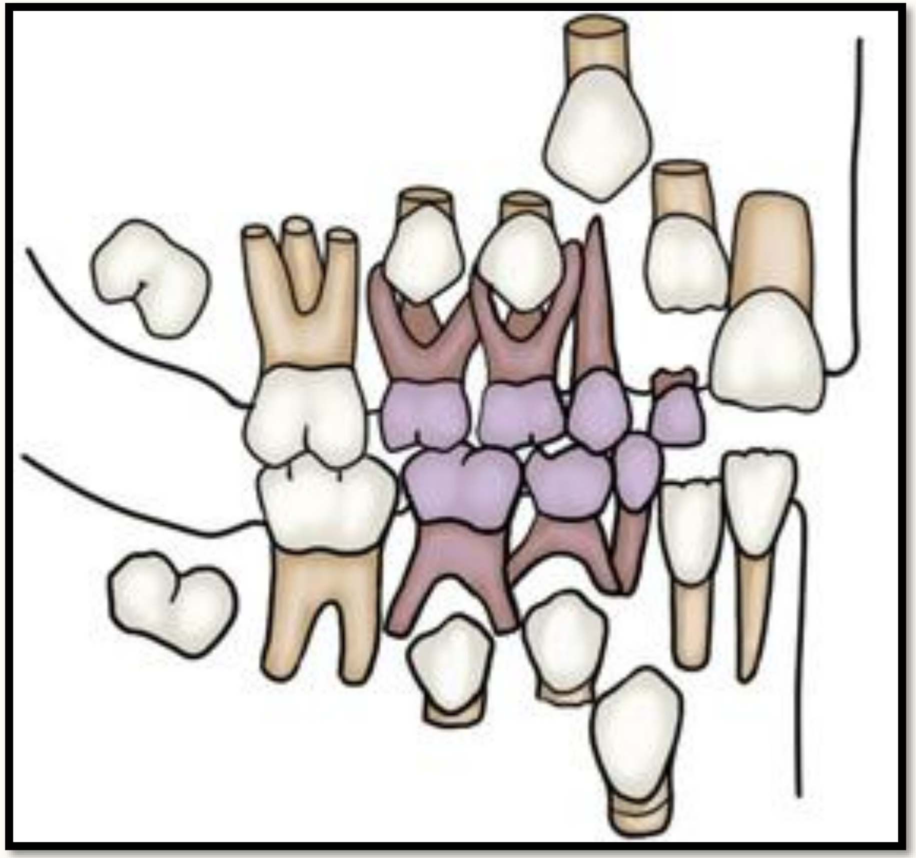
	Calcification date (Months)	Crown completed	Eruption (Years)
1/1	3-4	- 3 years before eruption	7/6
2/2	10-12 / 3-4		8/7
3/3	4-5		11/9
4/4	1½ / 1¾	- 3 years before eruption	10
5/5	2 / 2¼		11-12
6/6	At birth	-3-4 years before eruption	6
7/7	3 years		12
8/8	8 years		18



6-7 YEARS

6 years due to eruption of upper 6 – 7 years till upper 1 eruption and upper A exfoliation.

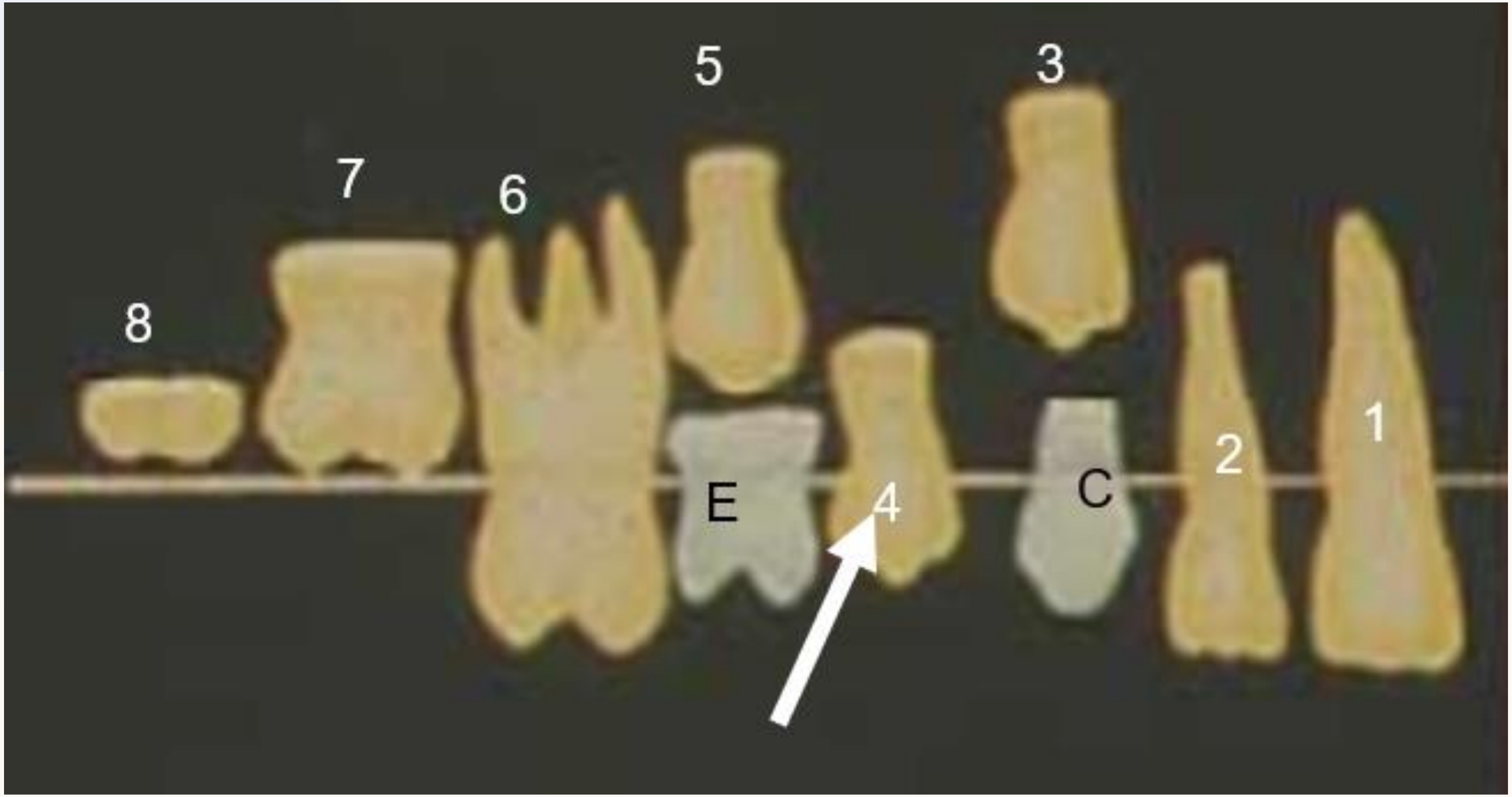
	Calcification date (Months)	Crown completed	Eruption (Years)
1/1	3-4	- 3 years before eruption	7/6
2/2	10-12/3-4		8/7
3/3	4-5		11/9
4/4	$1\frac{1}{2}$/$1\frac{3}{4}$	- 3 years before eruption	10
5/5	2/$2\frac{1}{4}$		11-12
6/6	At birth	-3-4 years before eruption	6
7/7	3 years		12
8/8	8 years		18



7-8 YEARS

Due to eruption of upper 1 (and crown completion of upper 4) at 7 years –8 years till eruption of upper 2 and (beginning of calcification of upper 8).

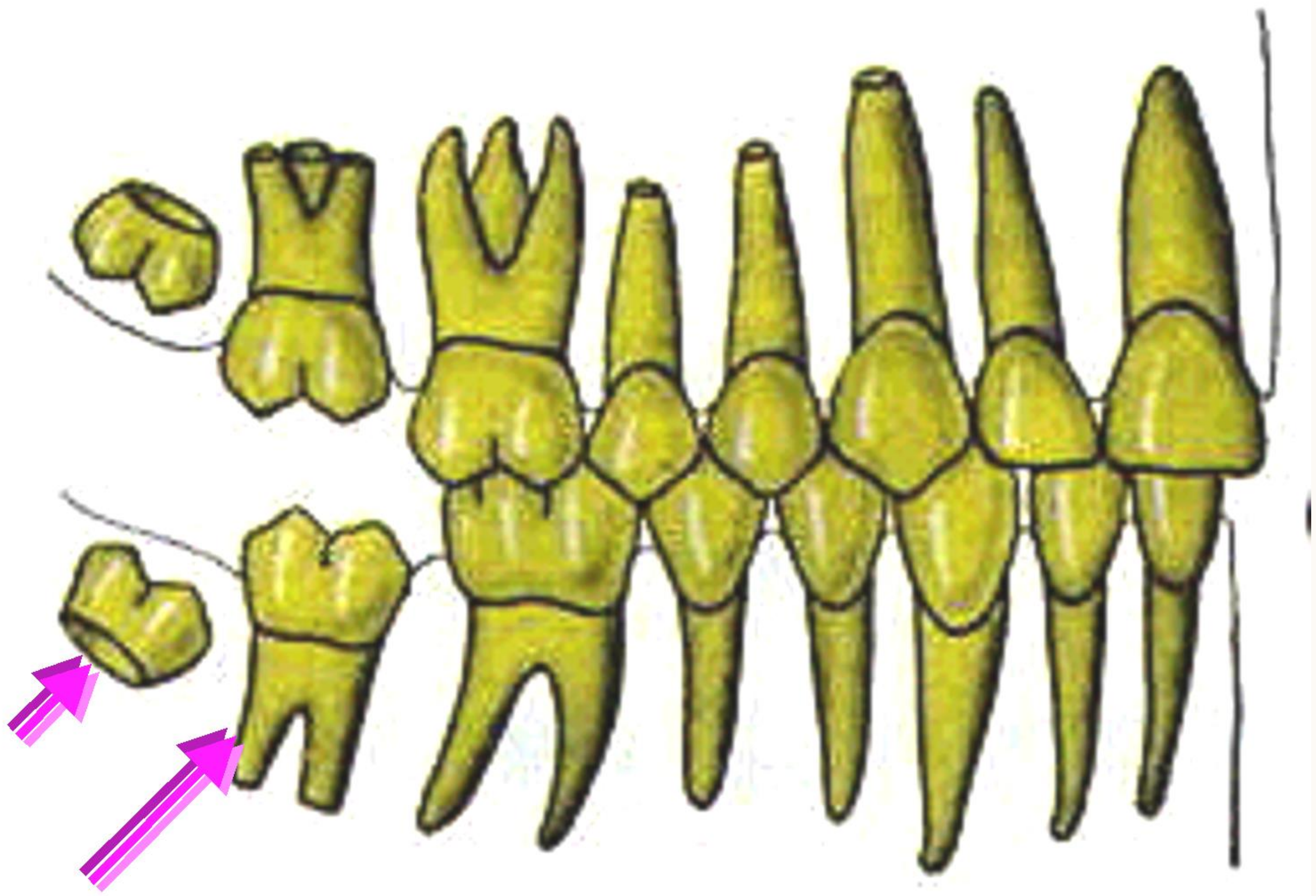
	Calcification date (Months)	Crown completed	Eruption (Years)
1/1	3-4	- 3 years before eruption	7/6
2/2	10-12 / 3-4		8/7
3/3	4-5		11/9
4/4	$1\frac{1}{2}$ / $1\frac{3}{4}$	- 3 years before eruption	10
5/5	2 / $2\frac{1}{4}$		11-12
6/6	At birth	-3-4 years before eruption	6
7/7	3 years		12
8/8	8 years		18



10 years due to eruption of upper 4 – 11 years till eruption of upper 3 and 5.

10-11 YEARS

	Calcification date (Months)	Crown completed	Eruption (Years)
1/1	3-4	- 3 years before eruption	7/6
2/2	10-12/3-4		8/7
3/3	4-5		11/9
4/4	$1\frac{1}{2}$/$1\frac{3}{4}$	- 3 years before eruption	10
5/5	2/$2\frac{1}{4}$		11-12
6/6	At birth	-3-4 years before eruption	6
7/7	3 years		12
8/8	8 years		18

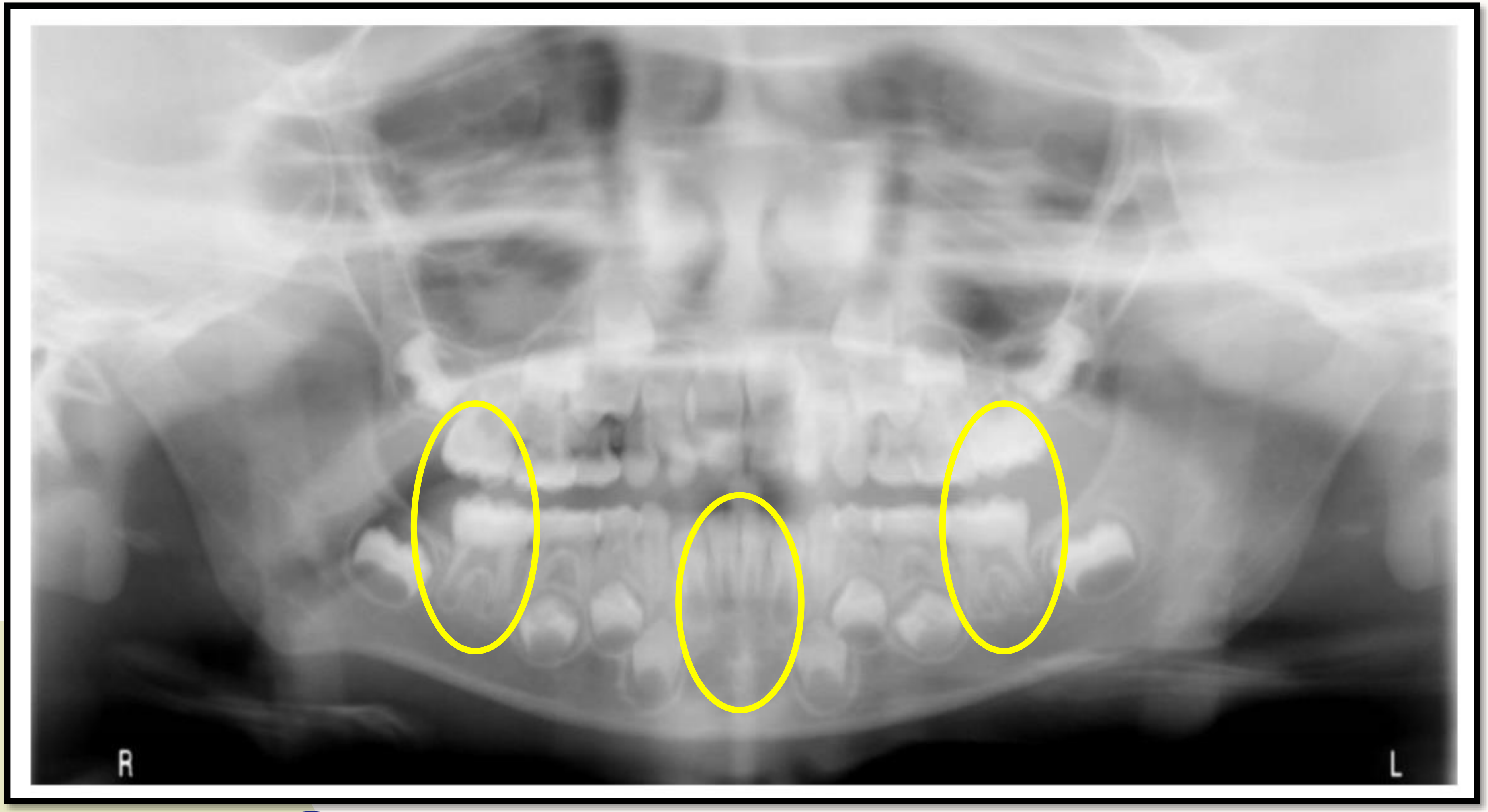


Since 12 years is the eruption of 7-
And 13 years root of 4 completes.

12-13 YEARS

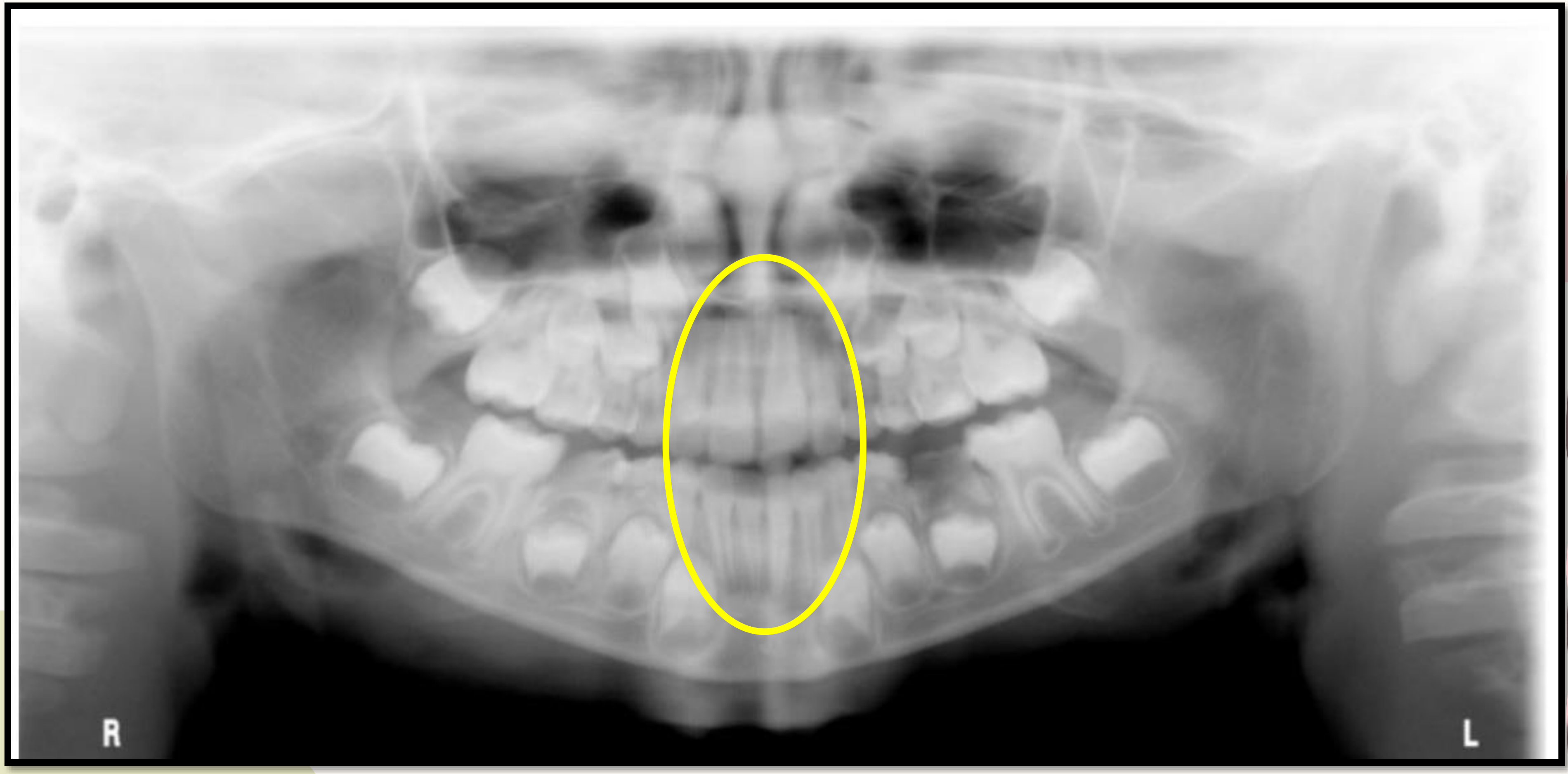
Student Self-Assessment

6-7 YEARS



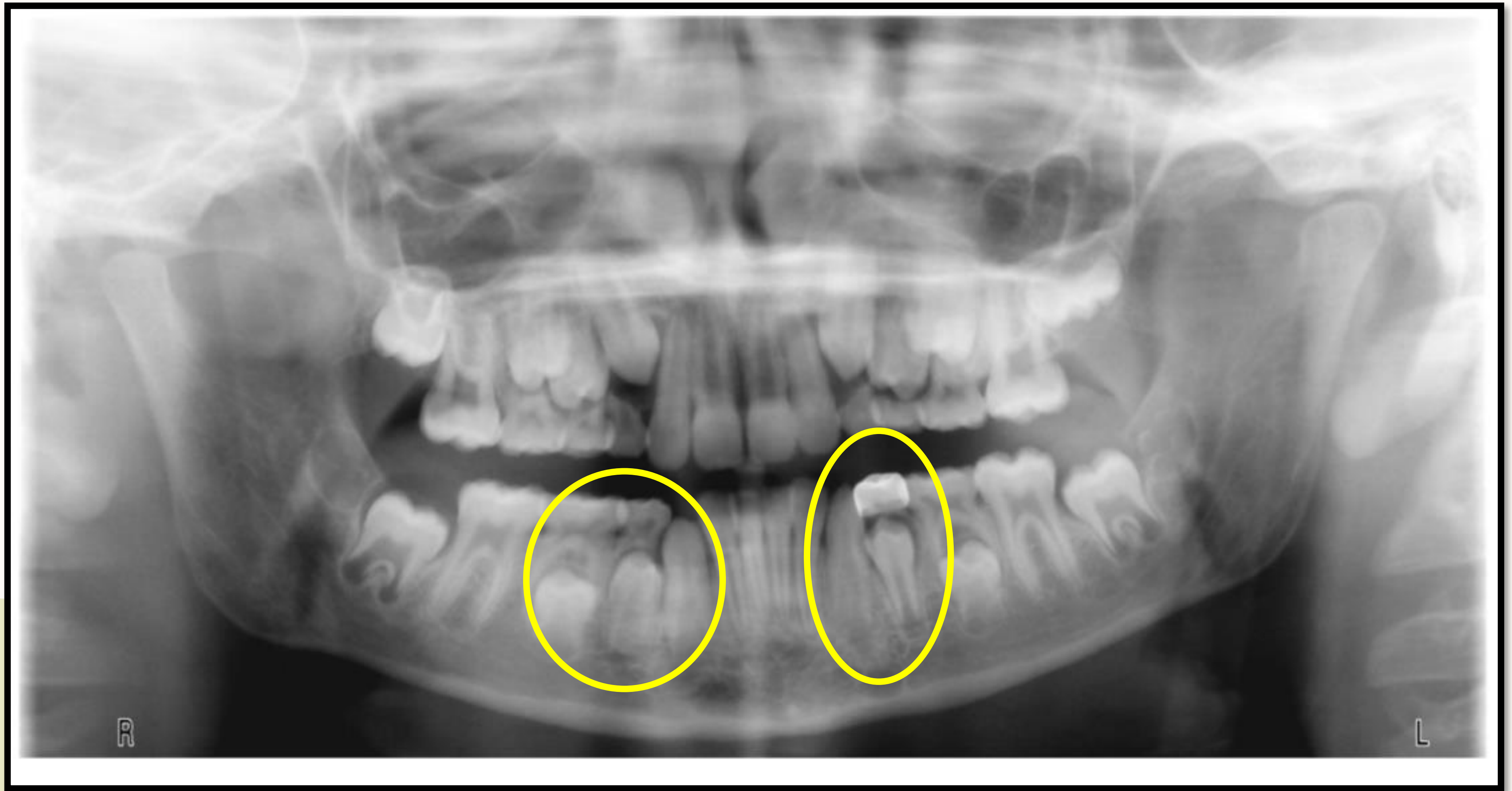
Eruption of permanent molars and mandibular incisors.

8-9 YEARS



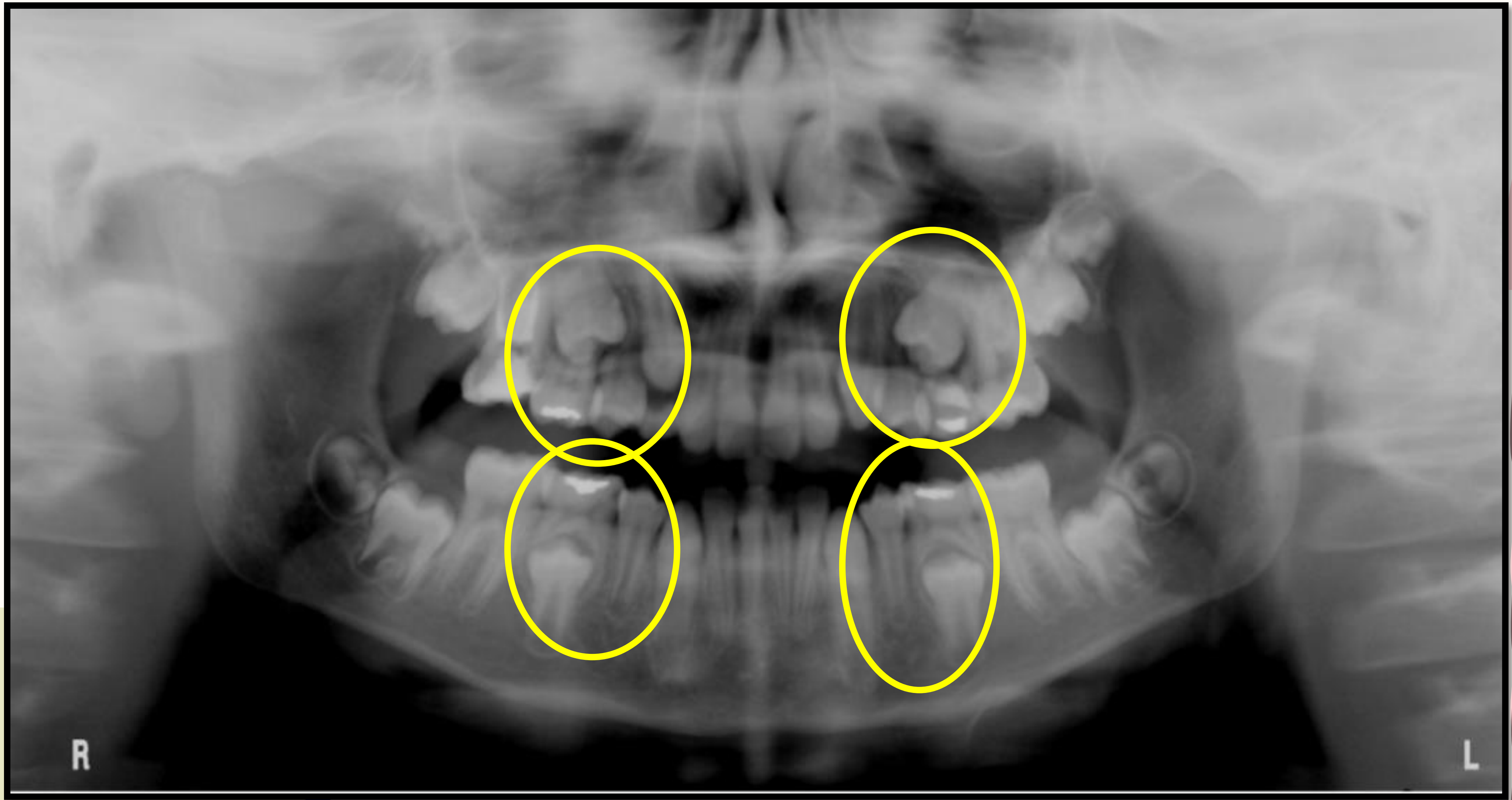
Eruption of permanent incisors, while mandibular canines not yet.

9-10 YEARS



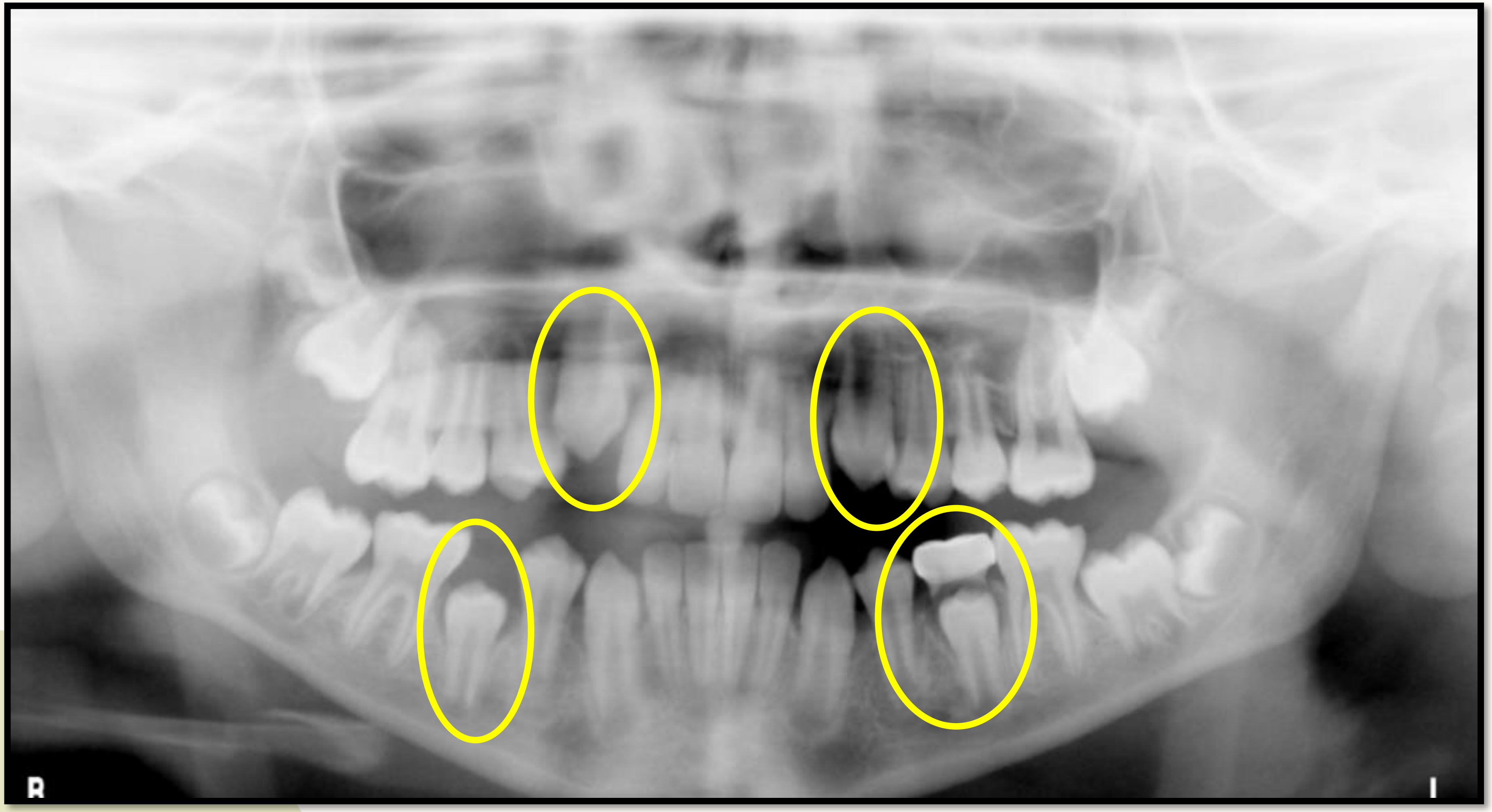
Eruption of mandibular canines, while premolars not yet.

10-11 YEARS



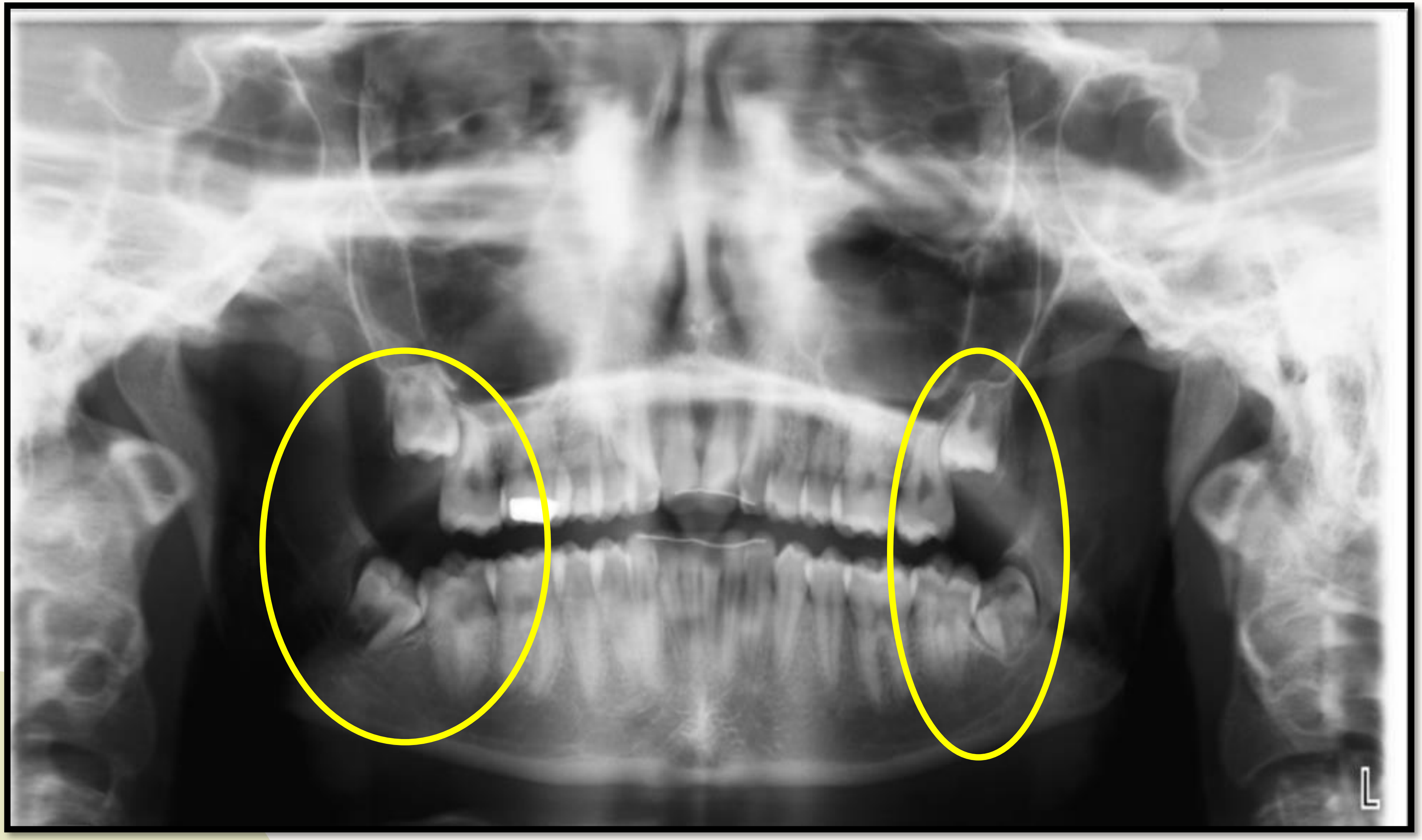
Eruption of 1st premolars, but maxillary 3 and 5 not yet.

11-12 YEARS



Eruption of 2nd premolars and upper canines, but 2nd molars not yet.

15-18 YEARS



Closure of 2nd molars roots, but 3rd molars not yet erupted.

Choose the appropriate dental age shown on this radiograph?

- 7-9 years old
- 10-12 years old
- 13-17 years old
- >17 years old

This baby in this picture is about 6-8 months. True or False?

- [x] True
- [ ] False

How old is this boy?

- ○ 3-4 years old
- ● 6-7 years old
- ○ 8-9 years old
- ○ 10-12 years old


THANK YOU!